Direction of vorticity on the high set: what is closed, what is false, and the remaining Campanato bound

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Abstract

This note records, as a closed ledger, the geometric structure of vorticity direction for unmodified 3D Navier–Stokes on the maximal classical interval $[0, T^*)$. Write $\omega = \nabla \times u$, $r = |\omega|$, $\xi = \omega/r$ on $\{r > 0\}$, $S = \frac{1}{2}(\nabla u + (\nabla u)^\top)$. Three facts are established: (i) the implication from uniform $\frac{1}{2}$-Hölder continuity of ξ on $\{r \geq \Omega\}$ (Definition 7.1) to $T^* = \infty$, by the Constantin–Fefferman kernel and Beirão da Veiga–Berselli stretching, HLS, absorption, Gronwall, and Kato H^1; (ii) the exact direction equation on $\{r > 0\}$, with projected drift and quadratic tension $\nu|\nabla\xi|^2\xi$; (iii) $S \in L^2_t L^2_x$ from energy and Calderón–Zygmund on L^2, with no $\|\omega\|_{L^\infty}$. Two claims are discharged as false: the drift is not $O(1/\Omega)$; De Giorgi–Nash–Moser does not yield uniform $C^{0,1/2}$ on the moving high set $E(t)$. Definition 7.1 itself is not proved. Energy does not prove it. The remaining statement is the Campanato bound on $4\sin^2(\phi/2)$ on $E(t)$. This note does not assert a solution of the Clay problem.

MSC-style keywords: Navier–Stokes, vorticity direction, Beale–Kato–Majda, Constantin–Fefferman, Beirão da Veiga–Berselli.

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*No tether. Degeneracy of $\xi = \omega/|\omega|$ is respected. This note is the canonical ledger of the geometric criterion: the implication 7.1 $\Rightarrow T^* = \infty$ is closed; Definition 7.1 itself is not. It does not claim a solution of the Clay problem.

1 Canonical ledger

What this note discovers is not a global regularity theorem. It is the precise partition of a geometric criterion into closed identities, false shortcuts, and one remaining estimate.

Piece	Status
7.1 $\Rightarrow$ high–high kernel $\lvert y\rvert^{-5/2}$, HLS, absorb, Gronwall, Kato H^1, Serrin	Closed as an implication (CF / Beirão da Veiga–Berselli)
ξ-equation on $\{r > 0\}$	Closed, with P_ξ and $\lvert\nabla\xi\rvert^2\xi$
$S \in L^2_t L^2_x$ from energy + CZ on L^2	Closed, no $\lvert\omega\rvert_\infty$
Drift $= O(1/\Omega)$	False
DGNM $\Rightarrow$ uniform $C^{0,1/2}$ on $E(t)$	False (system, $p = 2 < 5/2$, free boundary)
Definition 7.1 itself	Not proved

The only thing left is Definition 7.1, from the direction equation on $\{r \geq \Omega\}$, without $\|\omega\|_{L^\infty}$. The mean oscillation $\operatorname{osc}_{B_\rho}\xi \leq C(\Omega,\nu)\,\rho^{1/2}$ *is* Definition 7.1. Interior estimates miss $\partial E(t)$. Energy, HLS on the high–high piece, absorption, Gronwall, Kato H^1, and Serrin are the implication 7.1 $\Rightarrow T^* = \infty$. They are not a proof of 7.1.

2 Scope

This note treats the unmodified Cauchy problem for 3D incompressible Navier–Stokes, written as (1)–(2) below, on the maximal classical existence interval $[0, T^*)$. The only geometric input is a uniform spatial $\frac{1}{2}$-Hölder bound on the vorticity *direction* on the set where the vorticity *magnitude* is at least a positive threshold. From that input one concludes $T^* = \infty$.

The following are **not** proved: that $C(t)$ remains finite up to T^* for every smooth divergence-free u_0; that $T^* < \infty$ occurs for some such u_0; that a named tether or metriplectic correction is present in the equations. The PDE used is (1).

Section 10 constructs candidate constants from the PDE and tests whether the energy identity forces them to stay finite. It does not.

3 The equations

Let $\nu > 0$. On $\mathbb{T}^3 = \mathbb{R}^3/(2\pi\mathbb{Z})^3$ (the whole-space case is identical up to replacing the periodized Newtonian potential by $1/(4\pi|x|)$) one considers

$$\partial_t u + (u \cdot \nabla)u = \nu\Delta u - \nabla p, \tag{1}$$

$$\nabla \cdot u = 0, \tag{2}$$

with $u(\,\cdot\,,0) = u_0 \in C^\infty(\mathbb{T}^3;\mathbb{R}^3)$, $\nabla \cdot u_0 = 0$. Local well-posedness in H^s, $s > 5/2$, yields a unique classical solution on a maximal interval $[0,T^*)$, $T^* \in (0,\infty]$, with the standard blow-up alternative: if $T^* < \infty$ then

$$\limsup_{t\uparrow T^*} \|\nabla u(t)\|_{L^2(\mathbb{T}^3)} = \infty. \tag{3}$$

Equivalently, $\limsup_{t\uparrow T^*} \|\omega(t)\|_{L^2} = \infty$, since $\|\omega\|_{L^2} = \|\nabla u\|_{L^2}$ under (2). This is the energy-class continuation criterion (Kato); it is the only continuation tool used below. (Beale–Kato–Majda in $L^1_t L^\infty_x$ is stronger than needed once enstrophy is bounded.)

Curl of (1), using (2), gives the vorticity equation

$$\partial_t\omega + (u \cdot \nabla)\omega = (\omega \cdot \nabla)u + \nu\Delta\omega, \qquad \omega = \nabla \times u. \tag{4}$$

Write $r := |\omega|$ and, on the open set $\{r > 0\}$,

$$\xi := \frac{\omega}{r}. \tag{5}$$

The strain is the symmetric gradient

$$S := \frac{1}{2}\big(\nabla u + (\nabla u)^\top\big). \tag{6}$$

On $\{r > 0\}$ one has the stretching identity

$$(\omega \cdot \nabla)u \cdot \omega = r^2\,(S\xi \cdot \xi). \tag{7}$$

The left-hand side of (7) extends continuously by 0 at $r = 0$. The unit field ξ is *not* extended; every estimate below that mentions $\xi(x)$ is restricted to $r(x) > 0$.

4 The two constants

The geometric criterion uses exactly two solution-class constants, both allowed to depend on the data (u_0,ν) and on nothing else from the later evolution:

$$\Omega = \Omega(u_0,\nu) > 0, \tag{8}$$

$$C = C(u_0,\nu) < \infty. \tag{9}$$

Here Ω is the vorticity-intensity threshold: ξ is constrained only on $\{r \geq \Omega\}$. The complementary set $\{r < \Omega\}$ is estimated without forming ξ. The constant C is the uniform spatial $\frac{1}{2}$-Hölder modulus of ξ on that high set. From (8)–(9) the enstrophy argument produces one derived majorant coefficient, depending on the two constants and on viscosity,

$$K(C,\Omega,\nu) = \frac{C_*}{\nu}\,(C+\Omega)^2, \tag{10}$$

where $C_* < \infty$ depends only on the Sobolev, Hardy–Littlewood–Sobolev, and Calderón–Zygmund constants of $\mathbb{T}^3$ (or $\mathbb{R}^3$). The pair (Ω, C) is the input; K is not an independent hypothesis.

These two constants are not constructed from (1) in this note. Theorem A takes (8)–(9) as given.

5 The criterion

Theorem A (Beirão da Veiga–Berselli, recast)**.** *Let u be the classical solution of* (1)–(2) *on* $[0, T^*)$*, and let* $\Omega = \Omega(u_0, \nu)$ *and* $C = C(u_0, \nu)$ *be the two constants* (8)–(9)*. Suppose that for all* $t \in [0, T^*)$ *and all* $x, y \in \mathbb{T}^3$ *with* $r(x,t) \geq \Omega$ *and* $r(y,t) \geq \Omega$*,*

$$|\xi(x,t) - \xi(y,t)| \leq C\,|x-y|^{1/2}. \tag{11}$$

Then $T^* = \infty$*, and* $u \in C^\infty(\mathbb{T}^3 \times [0,\infty))$*. The continuation bound depends on the two constants only through* $K(C, \Omega, \nu)$ *in* (10)*:*

$$\sup_{t<T^*} \|\omega(t)\|_{L^2}^2 \leq \|\omega(0)\|_{L^2}^2 \exp\Big(\frac{K(C,\Omega,\nu)}{2\nu}\,\|u_0\|_{L^2}^2\Big). \tag{12}$$

The Hölder condition may be localized to $|x-y| < \delta$ for an arbitrary $\delta > 0$; the far field is controlled by the integrable periodized kernel. Only pairs in which *both* magnitudes meet the threshold Ω are constrained. Pairs with $r(x) < \Omega$ or $r(y) < \Omega$ are estimated crudely, without using ξ.

Remark 1. On $\{r(x), r(y) > 0\}$ one has $|\xi(x) - \xi(y)| = 2|\sin(\theta/2)|$ and $\sin\theta \leq |\xi(x) - \xi(y)|$, where $\theta = \angle(\omega(x), \omega(y))$. Thus (11) implies $\sin\theta(x,y,t) \leq C|x-y|^{1/2}$ on the high-high set, which is Assumption A of Beirão da Veiga–Berselli [1] at exponent $\alpha = 1/2$ with g a constant. Lipschitz direction (Constantin–Fefferman [3]) is the case $\alpha = 1$.

6 Enstrophy and stretching

Take the L^2 inner product of (4) with ω. After an integration by parts (justified on every compact subinterval of $[0, T^*)$ by classical smoothness),

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|\omega\|_{L^2}^2 + \nu\|\nabla\omega\|_{L^2}^2 = \int_{\mathbb{T}^3} \mathcal{K}(x,t)\,\mathrm{d}x, \tag{13}$$

where the stretching density is

$$\mathcal{K}(x,t) := \big((\omega\cdot\nabla)u\cdot\omega\big)(x,t). \tag{14}$$

On $\{r = 0\}$ one has $\mathcal{K} = 0$. On $\{r > 0\}$ one has $\mathcal{K} = r^2(S\xi\cdot\xi)$.

Biot–Savart on $\mathbb{T}^3$ reconstructs u from ω by the periodized Newtonian potential G. Differentiating under the principal value gives S as a Calderón–Zygmund operator of ω. The Constantin–Fefferman representation [3] then yields the pointwise bound, for a dimensional constant c_0 independent of u,

$$|\mathcal{K}(x)| \leq c_0 \int_{\mathbb{T}^3} \frac{r(x)^2\, r(y)\, \sin\theta(x,y)}{|x-y|^3}\,\mathrm{d}y, \tag{15}$$

where the integrand is read as 0 if $r(x) = 0$ or $r(y) = 0$, so that ξ is never invoked on the degeneracy set. (On the torus, $|x-y|$ denotes chordal/periodized distance; the local singularity is the same as in $\mathbb{R}^3$.)

7 High-high versus complementary pairs

Fix $\Omega > 0$ as in Theorem A. Split the integral in (15) according to whether $r(y) \geq \Omega$ or not. Write $\mathcal{K} = \mathcal{K}_{\mathrm{hh}} + \mathcal{K}_{\mathrm{c}}$.

High-high. If $r(x) \geq \Omega$ and $r(y) \geq \Omega$, (11) and $\sin\theta \leq |\xi(x) - \xi(y)|$ give $\sin\theta/|x-y|^3 \leq C|x-y|^{-5/2}$. Hence

$$|\mathcal{K}_{\mathrm{hh}}(x)| \leq c_0 C\, r(x)^2\, I(x), \qquad I(x) := \int_{\mathbb{T}^3} \frac{r(y)}{|x-y|^{5/2}}\,\mathrm{d}y. \tag{16}$$

The field I is a Riesz potential of order $\beta = 1/2$ applied to r. Hardy–Littlewood–Sobolev on $\mathbb{T}^3$ (or $\mathbb{R}^3$) yields

$$\|I\|_{L^3} \leq C_{\mathrm{HLS}} \|\omega\|_{L^2}, \tag{17}$$

because $1/3 = 1/2 - (1/2)/3$. If $r(x) < \Omega$, the factor $r(x)^2$ already places $\mathcal{K}_{\mathrm{hh}}(x)$ in the complementary class treated next; alternatively one may restrict (16) to $\{r(x) \geq \Omega\}$ and estimate the remainder with $\sin\theta \leq 1$ and $r(x) < \Omega$.

Complementary pairs. If $r(y) < \Omega$, use $\sin\theta \leq 1$ and do *not* form $\xi(y)$. After the usual cancellation in the CZ kernel of S (mean-zero homogeneous kernel of degree -3), the complementary stretching is controlled by the Hardy–Littlewood maximal function of the truncated vorticity:

$$|\mathcal{K}_{\mathrm{c}}(x)| \leq c_1 r(x)^2\, M\big(r\, 1_{\{r<\Omega\}}\big)(x). \tag{18}$$

In particular, using $r1_{\{r<\Omega\}} \leq \Omega$ and the boundedness of $M : L^2 \to L^2$,

$$\int_{\mathbb{T}^3} |\mathcal{K}_{\mathrm{c}}|\,\mathrm{d}x \leq c_2 \Omega\, \|\omega\|_{L^2} \|\omega\|_{L^6}. \tag{19}$$

(The same bound follows from $|S(1_{\{r<\Omega\}}\omega)| \lesssim M(r1_{\{r<\Omega\}})$. No division by r occurs.)

8 Closing the enstrophy inequality

Combining (13), (16), (17) and (19),

$$\int |\mathcal{K}_{\mathrm{hh}}|\,\mathrm{d}x \leq c_0 C \int r^2 I\,\mathrm{d}x \leq c_0 C \|\omega\|_{L^3}^2 \|I\|_{L^3} \leq c_3 C \|\omega\|_{L^2} \|\omega\|_{L^6} \|\omega\|_{L^2},$$

where the interpolation $\|\omega\|_{L^3}^2 \leq \|\omega\|_{L^2}\|\omega\|_{L^6}$ was used. Thus

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|\omega\|_{L^2}^2 + \nu \|\nabla\omega\|_{L^2}^2 \leq c_4 (C + \Omega)\, \|\omega\|_{L^2}^2 \|\omega\|_{L^6}. \tag{20}$$

Sobolev on $\mathbb{T}^3$ gives $\|\omega\|_{L^6} \leq C_{\mathrm{S}} \|\nabla\omega\|_{L^2}$. Young's inequality, for every $\varepsilon > 0$,

$$\|\omega\|_{L^2}^2 \|\omega\|_{L^6} \leq \frac{\varepsilon}{c_4(C+\Omega)} \|\omega\|_{L^6}^2 + \frac{c_4(C+\Omega)}{4\varepsilon} \|\omega\|_{L^2}^4,$$

so choosing ε small enough to absorb the first term into $\nu\|\nabla\omega\|_{L^2}^2$ produces

$$\frac{\mathrm{d}}{\mathrm{d}t}\|\omega\|_{L^2}^2 + \nu \|\nabla\omega\|_{L^2}^2 \leq K(C, \Omega, \nu)\, \|\omega\|_{L^2}^4, \tag{21}$$

with $K(C, \Omega, \nu)$ the derived coefficient (10). The only data-dependent constants entering K are the two constants Ω and C.

9 Continuation

Energy equality for (1) on $[0,T^*)$ yields

$$\frac{1}{2}\|u(t)\|_{L^2}^2 + \nu\int_0^t \|\nabla u(s)\|_{L^2}^2\,\mathrm{d}s = \frac{1}{2}\|u_0\|_{L^2}^2, \tag{22}$$

hence $\int_0^t \|\omega(s)\|_{L^2}^2\,\mathrm{d}s \le (2\nu)^{-1}\|u_0\|_{L^2}^2$ for every $t < T^*$. Drop the dissipation in (21) and apply Gronwall:

$$\|\omega(t)\|_{L^2}^2 \le \|\omega(0)\|_{L^2}^2 \exp\Big(K\int_0^t \|\omega(s)\|_{L^2}^2\,\mathrm{d}s\Big) \le \|\omega(0)\|_{L^2}^2 \exp\big(K(2\nu)^{-1}\|u_0\|_{L^2}^2\big). \tag{23}$$

The right-hand side is independent of $t < T^*$. Therefore $\sup_{t<T^*}\|\nabla u(t)\|_{L^2} < \infty$. The blow-up alternative (3) forces $T^* = \infty$.

Once the solution exists globally in H^1 with uniformly bounded enstrophy, standard parabolic bootstrapping for (1) (Kato, or L^p-L^q estimates for the Stokes semigroup) upgrades u to $C^\infty(\mathbb{T}^3 \times [0,\infty))$.

This is Theorem A.

10 Construction from the PDE, and the test

The two constants are now built from (1)–(2) and from u_0, then tested against the energy identity. No term is added to the PDE. Degeneracy of ξ on $\{r = 0\}$ is respected.

10.1 Irrotational data

If $\omega_0 \equiv 0$, then u_0 is a harmonic divergence-free field on $\mathbb{T}^3$, hence constant. The solution of (1)–(2) is $u(\,\cdot\,, t) = u_0$, which is smooth for all time. Take $\Omega = 1$ and $C = 0$; the high set is empty. This case is settled and is excluded below.

10.2 The threshold Ω

Fix the intensity threshold independently of the later evolution,

$$\Omega(u_0, \nu) := 1. \tag{24}$$

Any other fixed positive number may replace 1; Beirão da Veiga–Berselli allow an arbitrary $k > 0$. The choice must *not* track $\|\omega(t)\|_\infty$. In particular one cannot take $\Omega(t) > \|\omega(t)\|_\infty$, which would empty the high set and make (11) vacuous by circular use of a bound on ω. The value (24) depends on (u_0, ν) only in that $\nu > 0$ and u_0 is given; it is a universal positive threshold.

10.3 The modulus $C(t)$ along the classical solution

Let $E(t) := \{x \in \mathbb{T}^3 : r(x,t) \ge \Omega\}$. On each compact subinterval of $[0,T^*)$ the solution is smooth, so $E(t)$ is closed and $\xi(t)$ is smooth on a neighbourhood of $E(t)$ whenever $E(t)$ is nonempty (because $r \ge 1$ there). Define the running $\frac{1}{2}$-Hölder modulus on the high set by

$$C(t) := \begin{cases} 0 & \text{if } E(t) = \emptyset, \\ \sup\big\{\frac{|\xi(x,t)-\xi(y,t)|}{|x-y|^{1/2}} : x, y \in E(t),\ x \ne y\big\} & \text{if } E(t) \ne \emptyset. \end{cases} \tag{25}$$

Local smoothness gives $C(t) < \infty$ for every $t < T^*$. The constant required by Theorem A is the *uniform* value

$$C(u_0, \nu) := \sup_{t\in[0,T^*)} C(t) \in [0, \infty]. \tag{26}$$

This is constructed from the PDE in the only non-circular sense available: it is the actual modulus of the classical solution on its maximal interval. Theorem A applies if and only if this supremum is finite.

10.4 Direction equation on $\{r > 0\}$

On $\{r > 0\}$, $|\xi| = 1$ and $(\omega \cdot \nabla)u = S\omega$. The chain rule in (4) yields

$$D_t\xi = P_\xi(S\xi) + \nu\, r^{-1} P_\xi(\Delta\omega), \tag{27}$$

where $D_t = \partial_t + u \cdot \nabla$ and $P_\xi v := v - (\xi \cdot v)\xi$. Expand $\Delta\omega = \Delta(r\xi)$:

$$\Delta\omega = (\Delta r)\xi + 2\nabla r \cdot \nabla\xi + r\Delta\xi.$$

Then $P_\xi(\Delta\omega) = 2P_\xi(\nabla r \cdot \nabla\xi) + rP_\xi(\Delta\xi)$. The constraint $|\xi| = 1$ gives $\xi \cdot \nabla\xi = 0$ and $\xi \cdot \Delta\xi = -|\nabla\xi|^2$, hence $P_\xi(\Delta\xi) = \Delta\xi + |\nabla\xi|^2\xi$. Substituting produces the correct viscous equation

$$D_t\xi - \nu\Delta\xi = P_\xi(S\xi) + 2\nu r^{-1} P_\xi(\nabla r \cdot \nabla\xi) + \nu|\nabla\xi|^2\xi. \tag{28}$$

The drift is projected: it is $P_\xi(\nabla r \cdot \nabla\xi)$, not $\nabla r \cdot \nabla\xi$. The quadratic term $\nu|\nabla\xi|^2\xi$ is the second fundamental form of S^2 (harmonic-map tension) and must be kept. Componentwise $\Delta\xi + 2\nabla \log r \cdot \nabla\xi = r^{-2}\mathrm{div}(r^2\nabla\xi)$, but the equation uses the projection of that operator, not the unprojected drift.

On $E(t) = \{r \geq \Omega\}$ one has $|\nabla\xi| \leq \Omega^{-1}|\nabla\omega|$ and $|\nabla r| \leq |\nabla\omega|$ (Pythagoras $|\nabla\omega|^2 = |\nabla r|^2 + r^2|\nabla\xi|^2$). In particular $|\nabla r| \leq r$ is false, and $|\nabla \log r| \leq \Omega^{-1}|\nabla\omega|$ is not a bound independent of the solution: distance to $\partial E(t)$ does not control $|\nabla r|$ inside $E(t)$. The drift is therefore not $O(1/\Omega)$ in any uniform sense. No division by r occurs on $\{r = 0\}$.

10.5 Morrey reduction

Morrey's embedding on $\mathbb{T}^3$ gives $W^{1,6} \hookrightarrow C^{0,1/2}$. Let $\chi \in C^\infty(\mathbb{R})$ satisfy $\chi(s) = 0$ for $s \leq 1$ and $\chi(s) = 1$ for $s \geq 2$, and set $v(x,t) := \chi\big(r(x,t)/\Omega\big)\,\xi(x,t)$ on $\{r > 0\}$, with $v = 0$ on $\{r = 0\}$. Then v is a globally defined smooth field on $[0, T^*)$, $v = \xi$ on $\{r \geq 2\Omega\}$, and

$$|\nabla v| \leq C_\chi\, \Omega^{-1}|\nabla\omega|. \tag{29}$$

Hence a finite bound

$$\sup_{t<T^*} \|\nabla\omega(t)\|_{L^6(\mathbb{T}^3)} < \infty \tag{30}$$

would imply $\sup_{t<T^*} C(t) < \infty$ after replacing Ω by 2Ω in Theorem A. Lipschitz control of ξ would require $\nabla\omega \in L^\infty_t L^\infty_x$, which is Beale–Kato–Majda and is strictly stronger than (30).

10.6 What the energy identity supplies

Energy equality (22) yields only

$$u \in L^\infty(0, T^*; L^2) \cap L^2(0, T^*; H^1), \qquad \omega \in L^2(0, T^*; L^2).$$

It does *not* yield $\nabla\omega \in L^\infty_t L^6$, nor even $\nabla\omega \in L^2_t L^2$. The latter would be an enstrophy bound, which is what Theorem A is designed to produce after C is known to be finite. Using (30) as a hypothesis to bound C is therefore circular at the energy level.

A differential inequality for $C(t)$ from (27) contains the stretching $P_\xi(S\xi)$ and the viscous term $\nu\Omega^{-1}P_\xi(\Delta\omega)$ on $E(t)$. The strain S is a Calderón–Zygmund image of ω. Neither term is controlled in $L^\infty_t L^6$ by $\|\omega\|_{L^2_t L^2_x}$ alone. The same obstruction persists if $\Omega = 1$ is replaced by any other fixed positive threshold depending only on $\|u_0\|_{H^s}$ and ν.

10.7 Result of the test

Theorem B (construction from the PDE does not close)**.** *Let $u_0 \in C^\infty(\mathbb{T}^3;\mathbb{R}^3)$ be divergence-free and $\nu > 0$, and let u be the classical solution of* (1)–(2) *on* $[0, T^*)$*. Define* Ω *by* (24) *and* $C(u_0, \nu)$ *by* (26)*. Then:*

1. *If $\omega_0 \equiv 0$, then $T^* = \infty$ and u is globally smooth.*
2. *If $\omega_0 \not\equiv 0$, then $C(t) < \infty$ for every $t < T^*$, but the energy identity for* (1) *does not imply $C(u_0, \nu) < \infty$. In particular it does not imply* (30)*.*
3. *Consequently Theorem A cannot be applied from the energy class alone. The test does not prove $T^* = \infty$, and it does not produce a finite-time singularity.*

Local smoothness on compact subintervals of $[0, T^*)$ always gives a finite modulus on each such interval. Uniformity up to T^* is the whole problem. Values $\beta < 1/2$ are not known to suffice [2].

No term was added to (1). Degeneracy of ξ on $\{r = 0\}$ was respected throughout. Global regularity for arbitrary smooth data is not obtained from energy.

11 The geometric implication: Definition 7.1 $\Rightarrow T^* = \infty$

This section is the Constantin–Fefferman / Beirão da Veiga–Berselli implication, written so that it closes. If Definition 7.1 holds for a given smooth divergence-free u_0, then $T^* = \infty$ for that data. Definition 7.1 is not proved here; it is Theorem C and Lemma 1(B). This implication is not a regularity theorem for arbitrary data.

Definition 7.1 (geometric criterion). There exist $\Omega = \Omega(u_0, \nu) > 0$ and $C = C(u_0, \nu) < \infty$ such that for all $t \in [0, T^*)$ and all x, y with $r(x, t) \geq \Omega$ and $r(y, t) \geq \Omega$,

$$|\xi(x, t) - \xi(y, t)| \leq C\,|x - y|^{1/2}.$$

In particular $\sin\phi \leq C|y|^{1/2}$ on that high-high set, where $\phi = \angle(\omega(x), \omega(x + y))$. The constants C, Ω do not depend on T^*.

Definition 7.2 (continuation). If $T^* < \infty$ then $\limsup_{t \uparrow T^*} \|\nabla u(t)\|_{L^2} = \infty$ (Kato H^1). Equivalently $\limsup_{t \uparrow T^*} \|\omega(t)\|_{L^2} = \infty$. Beale–Kato–Majda ($\int_0^{T^*} \|\omega\|_{L^\infty} = \infty$) is a stronger necessary condition for blow-up; it is not the criterion fed by an L^2 enstrophy bound. Once $T^* = \infty$, Beale–Kato–Majda holds automatically on every finite interval.

Theorem (geometric regularity). Let $u_0 \in C^\infty(\mathbb{T}^3;\mathbb{R}^3)$, $\nabla \cdot u_0 = 0$, $\nu > 0$, and let u be the unique classical solution of (1)–(2) on $[0, T^*)$. Assume Definition 7.1. Then $T^* = \infty$ and $u \in C^\infty([0, \infty) \times \mathbb{T}^3)$.

Proof. *1. Energy.*

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|u\|_{L^2}^2 + \nu\|\nabla u\|_{L^2}^2 = 0,$$

so $\|u(t)\|_{L^2} \leq \|u_0\|_{L^2}$ for $t < T^*$. On $\mathbb{T}^3$, $\|\omega\|_{L^2} \simeq \|\nabla u\|_{L^2}$.

2. Enstrophy identity. Curl of (1) and the L^2 pairing give

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|\omega\|_{L^2}^2 = \int_{\mathbb{T}^3} (\xi \cdot S\xi)\, r^2\,\mathrm{d}x - \nu\|\nabla\omega\|_{L^2}^2. \tag{31}$$

The stretching density is $\mathcal{K} = r^2(\xi \cdot S\xi)$, not $\xi \cdot S\xi$. On $\{r = 0\}$ one has $\mathcal{K} = 0$. Pythagoras: $|\nabla\omega|^2 = |\nabla r|^2 + r^2|\nabla\xi|^2$.

3. Strain as a geometric kernel. Biot–Savart gives the Constantin–Fefferman representation

$$(\xi \cdot S\xi)(x) = \frac{3}{4\pi}\,\text{P.V.}\int_{\mathbb{T}^3} D\big(\hat{y}, \xi(x), \xi(x+y)\big)\frac{r(x+y)}{|y|^3}\,\mathrm{d}y,$$

with $D(\hat{y}, \xi, \xi') = (\hat{y}\cdot\xi)\det(\hat{y}, \xi', \xi)$ and $|D| \le |\sin\phi(x, x+y)| \le |\xi(x) - \xi(x+y)|$. If $\xi(x) = \xi(x+y)$ then $D = 0$: aligned vorticity does not stretch. That is the geometry.

4. Stretching under Definition 7.1. Split $\int(\xi \cdot S\xi)r^2$ into three pieces.

Low vorticity ($r(x) < \Omega$). Then $r^2 < \Omega^2$, so

$$\Big|\int_{\{r<\Omega\}} (\omega\cdot\nabla)u\cdot\omega\,\mathrm{d}x\Big| \le \Omega^2\int|\nabla u| \le C_{\mathbb{T}}\Omega^2\|\omega\|_{L^2}.$$

No $\|\omega\|_{L^\infty}$. Degeneracy: ξ is not used on this set.

Far interactions ($|y| \ge 1$). The kernel is smooth. Integrating by parts against $\omega = \nabla\times u$, the far strain is a bounded operator on u. Energy from Step 1 gives

$$|I_{\text{far}}| \le K_{\text{far}}\|u_0\|_{L^2}\|\omega\|_{L^2}^2.$$

Near high-high ($|y| < 1$ and $r(x), r(x+y) \ge \Omega$). Definition 7.1 gives $|D| \le C|y|^{1/2}$, so the kernel is $|y|^{-5/2}$. In three dimensions

$$\int_{|y|<1}|y|^{-5/2}\,\mathrm{d}y = \int_0^1 \rho^{-1/2}\,\mathrm{d}\rho < \infty:$$

the singularity is integrable. Hölder 1/2 is critical for this integrability (exponent $< 1/2$ makes the radial integral diverge). The contribution is $\int r(x)^2 I(x)\,\mathrm{d}x$ with $I = r * |\cdot|^{-5/2}$. Hardy–Littlewood–Sobolev at order 1/2 yields $\|I\|_{L^3} \le C_{\text{HLS}}\|\omega\|_{L^2}$. Interpolation $\|\omega\|_{L^3}^2 \le \|\omega\|_{L^2}\|\omega\|_{L^6}$ then gives

$$\int r^2 I \le C_{\text{HLS}}C\,\|\omega\|_{L^2}^2\|\omega\|_{L^6}. \tag{32}$$

Sobolev $\|\omega\|_{L^6} \le C_{\text{S}}\|\nabla\omega\|_{L^2}$ and Young with $\varepsilon = \nu/2$ produce

$$c\,\|\omega\|_{L^2}^2\|\omega\|_{L^6} \le \frac{\nu}{2}\|\nabla\omega\|_{L^2}^2 + C_\nu(C)\,\|\omega\|_{L^2}^4.$$

The exponent 4 is the one that closes. A bound $\|\omega\|_{L^2}^{5/2}\|\nabla\omega\|_{L^2}^{1/2}$ would Young-close to $\|\omega\|_{L^2}^{10/3}$, and $y' \le Ky^{5/3}$ for $y = \|\omega\|_{L^2}^2$ can blow up in finite time; that interpolation does not close. The chain (32) gives $y' \le Ky^2$, which Gronwall controls from energy.

5. Enstrophy inequality. Summing the three pieces,

$$\frac{\mathrm{d}}{\mathrm{d}t}\|\omega\|_{L^2}^2 + \nu\|\nabla\omega\|_{L^2}^2 \le K(C, \Omega, \nu)\,\|\omega\|_{L^2}^4 + C'\Omega^2\|\omega\|_{L^2}, \tag{33}$$

with $K = C_*(C+\Omega)^2/\nu$ as in (10). The right-hand side is not a constant independent of ω. Energy supplies $\int_0^t\|\omega\|_{L^2}^2\,\mathrm{d}s \le (2\nu)^{-1}\|u_0\|_{L^2}^2$. Gronwall yields, for every $t < T^*$,

$$\|\omega(t)\|_{L^2}^2 \le \big(\|\omega_0\|_{L^2}^2 + C''\big)\exp\Big(\frac{K}{2\nu}\|u_0\|_{L^2}^2\Big).$$

The exponential does not depend on t or on T^*, because C and Ω in Definition 7.1 do not. Hence $\omega \in L^\infty(0, T^*; L^2) \cap L^2(0, T^*; H^1)$, i.e. $u \in L^\infty(0, T^*; H^1(\mathbb{T}^3))$.

6. Serrin class and continuation. On $\mathbb{T}^3$, $H^1 \hookrightarrow L^6$, so $u \in L^\infty(0, T^*; L^6)$. Prodi–Serrin–Ladyzhenskaya: a Leray–Hopf solution in $L_t^sL_x^q$ with $3/q + 2/s \le 1$ and $q > 3$ is smooth. Here

$q = 6$, $s = \infty$, and $3/6 + 2/\infty = 1/2 < 1$, which is subcritical. Thus if $T^* < \infty$, u is smooth on $[0, T^*]$. Kato local existence: a smooth H^s solution ($s > 5/2$) with finite H^s norm at time T^* extends to $T^* + \varepsilon$. That contradicts maximality. Therefore $T^* = \infty$.

This is Kato–Fujita continuation in H^1, not an application of Beale–Kato–Majda to an L^2 bound. An L^2 bound on ω does not control $\int \|\omega\|_{L^\infty}$. It does control $\|u\|_{H^1}$, which is the continuation class used here. Beale–Kato–Majda is then automatic on every finite interval.

7. Global smoothness. Parabolic regularity for (1) on $\mathbb{T}^3$: a solution that is smooth on every compact time interval, with $T^* = \infty$, is C^∞ for all $t \geq 0$.

Thus Definition 7.1 $\Rightarrow$ Definition 7.2 cannot fire $\Rightarrow$ $T^* = \infty$ $\Rightarrow$ global smoothness. Alignment depletes stretching ($D = 0$ when $\xi(x) = \xi(x + y)$). Hölder 1/2 makes the leftover kernel integrable at the origin and matches HLS with $\omega \in L^2$. The resulting bound on enstrophy is independent of T^* because C, Ω are. Bounded enstrophy is $L^\infty_t H^1$, a subcritical Serrin class. Smooth solutions in that class cannot stop at a finite T^*.

That implication is unconditional *once* Definition 7.1 holds for the given u_0. It is not a theorem that Definition 7.1 holds for every smooth divergence-free initial datum. That last statement is Theorem C and Lemma 1(B), and is not proved here.

12 What is left

Everything in Theorems A–B is done. One statement remains. If it is proved, Theorem A gives $T^* = \infty$ for that datum. If it fails for some u_0, this route does not decide blow-up. This note does not claim that Definition 7.1 holds in general.

12.1 The remaining statement

Theorem C (remaining; not proved)**.** *Let $u_0 \in C^\infty(\mathbb{T}^3; \mathbb{R}^3)$ be divergence-free, $\nu > 0$, and u the classical solution of* (1)–(2) *on* $[0, T^*)$. *Set $\Omega = 1$ as in* (24)*. Then*

$$C(u_0, \nu) := \sup_{t \in [0, T^*)} C(t) < \infty, \tag{34}$$

with $C(t)$ as in (25)*. The finite value may depend on (u_0, ν) only.*

That is the entire remainder for this route. Equivalently, writing $E(t) = \{r(\,\cdot\,, t) \geq 1\}$,

$$\sup_{t < T^*} \sup_{\substack{x, y \in E(t) \\ x \neq y}} \frac{|\xi(x, t) - \xi(y, t)|}{|x - y|^{1/2}} < \infty. \tag{35}$$

No other estimate is missing from Theorem A: local existence, the stretching representation, the high-high / complementary split, Hardy–Littlewood–Sobolev at order 1/2, absorption into $\nu \|\nabla\omega\|^2_{L^2}$, Gronwall against the energy integral, and the H^1 blow-up alternative are already closed.

12.2 A legitimate reduction, and two that are not

On $E(t)$ one has $r \geq 1$, hence

$$|\xi(x) - \xi(y)| \leq 2|\omega(x) - \omega(y)|. \tag{36}$$

Thus (34) follows from uniform $\frac{1}{2}$-Hölder continuity of ω itself on $E(t)$. That is still a statement only about intense vorticity, not about $\|\omega\|_\infty$.

The following are *not* the remaining problem, because each is already known to be equivalent to global regularity by continuation, and using any of them to bound C is circular:

- $\sup_{t<T^*} \|\omega(t)\|_{L^2} < \infty$ (Kato H^1),
- $\int_0^{T^*} \|\omega(t)\|_{L^\infty} \, \mathrm{d}t < \infty$ (Beale–Kato–Majda),
- $\nabla\omega \in L^\infty(0, T^*; L^6)$ (Morrey, (30)),
- $u \in L_t^p L_x^q$ with $2/p + 3/q = 1$, $q > 3$ (Serrin).

The point of (34) is that it is geometrically weaker than a bound on $|\omega|$: a vortex that intensifies while remaining spatially aligned can have large r and still small $|\xi(x) - \xi(y)|$. Theorem A is designed to turn that alignment into an enstrophy bound. Beirão da Veiga [2] argues that $\beta = 1/2$ is critical for this method; the remaining statement is not a minor calculation.

12.3 What the direction equation must yield

The only unused dynamical information is (27) on $E(t)$. For $x, y \in E(t)$,

$$\begin{aligned} D_t^x \xi(x) - D_t^y \xi(y) &= P_{\xi(x)}\big(S(x)\xi(x)\big) - P_{\xi(y)}\big(S(y)\xi(y)\big) \\ &\quad + \nu\Big(r(x)^{-1} P_{\xi(x)} \Delta\omega(x) - r(y)^{-1} P_{\xi(y)} \Delta\omega(y)\Big), \end{aligned} \tag{37}$$

where $D_t^x = \partial_t + u(x) \cdot \nabla_x$. The first line is the misalignment of strain; the second is the viscous increment of direction. To prove (34) it is necessary and sufficient to show that this difference, divided by $|x - y|^{1/2}$, remains bounded on $E(t) \times E(t)$ up to T^*, with a bound depending only on (u_0, ν). Degeneracy is already removed: $r(x), r(y) \geq 1$.

That difference estimate is the next (and last) step on this route.

12.4 Beginning of the estimate for Definition 7.1

Fix $\Omega = 1$. For $x, y \in E(t)$ the elementary identity

$$|\xi(x) - \xi(y)|^2 = 2\big(1 - \xi(x) \cdot \xi(y)\big) = 4 \sin^2(\phi/2) \tag{38}$$

identifies the Hölder modulus of ξ with the CF angle in the kernel D. Thus Definition 7.1 is exactly the statement that the geometric factor $|D|$ is $O(|y|^{1/2})$ uniformly on high-high pairs, with a constant independent of $t < T^*$.

On $E(t)$ one also has (36). Along any segment that remains in $\{r \geq 1/2\}$,

$$\xi(x) - \xi(y) = \int_0^1 (x - y) \cdot \nabla\xi\big(y + s(x - y)\big) \, \mathrm{d}s, \tag{39}$$

and $|\nabla\xi| \leq 2|\nabla\omega|$ there. Bounding the right-hand side by $|x - y| \, \|\nabla\omega\|_{L^\infty}$ recovers a Lipschitz estimate, which is Beale–Kato–Majda, hence circular. Definition 7.1 asks for one derivative less: a factor $|x - y|^{1/2}$ rather than $|x - y|$. The unused dissipation in Pythagoras,

$$|\nabla\omega|^2 = |\nabla r|^2 + r^2 |\nabla\xi|^2,$$

puts $|\nabla\xi|$ in L^2 only after enstrophy is bounded, which is again circular. The estimate that would close Definition 7.1 without a derivative too many is a Campanato bound on $E(t)$,

$$\sup_{x \in E(t)} \sup_{\rho \in (0,1)} \rho^{-4} \int_{B_\rho(x) \cap E(t)} |\xi(z) - \xi_{x,\rho}|^2 \, \mathrm{d}z \leq C(u_0, \nu)^2, \tag{40}$$

equivalent to (34) on the high set. The integrand is controlled by $\sin^2(\phi/2)$ via (38), hence by the same D that depletes stretching. Whether (40) follows from (27) and energy, with no bound on $\|\omega\|_{L^\infty}$, is the whole of Definition 7.1.

No other analytic ingredient is missing. Once (40) holds, Section 11 is unconditional and $T^* = \infty$.

12.5 Corrected De Giorgi–Nash–Moser attempt

A proposed parabolic De Giorgi–Nash–Moser / Campanato argument for Definition 7.1 is rewritten here with the necessary corrections. It does not prove Definition 7.1. The surviving facts are recorded, then the gaps.

Forcing from Calderón–Zygmund $2 \to 2$, without $\|\omega\|_{L^\infty}$. The pointwise bound $|S| \le C|\omega|$ requires $\|\omega\|_{L^\infty}$ and is not used. Energy for (1) gives $\omega \in L^2(0,T^*;L^2)$ (equivalently $\nabla u \in L^2_t L^2_x$). The strain is a Calderón–Zygmund operator of ω, hence $S \in L^2(0,T^*;L^2)$ by the $L^2 \to L^2$ bound, globally, with no pointwise kernel estimate. Then $|P_\xi(S\xi)| \le |S|$, so

$$P_\xi(S\xi) \in L^2(0,T^*;L^2(\mathbb{T}^3)). \tag{41}$$

That term of (28) is under control.

Drift is not absorbed by Leray–Hopf. On a space-time cylinder $Q \subset \{r \ge \Omega\}$,

$$|2\nu r^{-1} P_\xi(\nabla r \cdot \nabla \xi)| \le 2\nu\Omega^{-1}|\nabla r|\,|\nabla\xi| \le 2\nu\Omega^{-2}|\nabla\omega|^2,$$

using $|\nabla r| \le |\nabla\omega|$ and $|\nabla\xi| \le \Omega^{-1}|\nabla\omega|$. (The bound $|\nabla r| \le |\omega|$ is false: Pythagoras gives $|\nabla r| \le |\nabla\omega|$, not $|\nabla r| \le r$.) Thus the drift lies in $L^2_t L^2_x(Q)$ only if $\nabla\omega \in L^4(Q)$. Leray–Hopf energy does not provide $\nabla\omega \in L^2$, let alone L^4. Ladyzhenskaya interpolation of $|\omega|^2|\nabla\omega|^2$ against $|\nabla\omega|^4$ and $|\omega|^4$ likewise requires norms of ω that energy does not give. There is no energy inequality for ω in the Leray–Hopf class; that would be an enstrophy bound, which is the output of Definition 7.1 rather than an input. The drift term in (28) is therefore not known to lie in $L^2_t L^2_x$ on cylinders in $\{r \ge \Omega\}$.

Quadratic tension. The term $\nu|\nabla\xi|^2\xi$ satisfies $|\nabla\xi|^2 \le \Omega^{-2}|\nabla\omega|^2$ on $E(t)$. Membership in $L^2_t L^2_x$ would require $\nabla\xi \in L^4$, i.e. again $\nabla\omega \in L^4$. Intrinsically, this term is normal to $T_\xi S^2$. The tangential equation is

$$P_\xi(D_t\xi - \nu\Delta\xi) = P_\xi(S\xi) + 2\nu r^{-1} P_\xi(\nabla r \cdot \nabla\xi), \tag{42}$$

which is the harmonic-map heat flow into S^2 with an L^2 stretching force and an uncontrolled drift. Finite-time singularities of the harmonic-map heat flow from three-dimensional domains into S^2 are known; quadratic growth in $\nabla\xi$ is not a perturbation of a scalar De Giorgi–Nash–Moser equation.

Scalar versus system. De Giorgi–Nash–Moser Hölder theory is for scalar parabolic equations. ξ is S^2-valued. In three dimensions, elliptic or parabolic systems with measurable coefficients need not be Hölder. The quadratic term $\nu|\nabla\xi|^2\xi$ is the harmonic-map tension field; three-dimensional harmonic-map heat flow can blow up.

Integrability of the reaction. For a bounded solution of $\theta_t + u \cdot \nabla\theta - \nu\Delta\theta = f$ in three space dimensions, a Hölder modulus from f needs $f \in L^p_{t,x}$ with $p > (n+2)/2 = 5/2$. Energy gives $P_\xi(S\xi) \in L^2_{t,x}$. That is below $5/2$. So even the scalar heat equation with this forcing does not yield $C^{0,1/2}$.

Free boundary. Interior estimates hold on cylinders at a definite distance from $\{r = \Omega\}$. Definition 7.1 is for all pairs in $E(t)$, including points near $\partial E(t)$, where $r \downarrow \Omega$ and $\nabla \log r$ can be large. Campanato needs small ρ, inside $E(t)$. Mixing ρ with the vorticity threshold Ω is the wrong scale. Integrating an L^2 bound against the free-space heat kernel does not prove $\operatorname{osc}_{B_\rho}\xi \le C(\Omega,\nu)\,\rho^{1/2}$ on the moving set $E(t)$: that oscillation bound *is* Definition 7.1.

What would actually close Definition 7.1. Prove one of the following from (28) on $\{r \geq \Omega\}$, with constants depending only on (u_0, ν, Ω), not on $\|\omega\|_{L^\infty}$ and not on $\|\nabla\omega\|_{L^2}$.

Lemma 1 (remaining; either form)**.** *(A) Integrability upgrade. On every cylinder at distance $\geq d(\Omega)$ from $\{r < \Omega\}$,*

$$P_\xi(S\xi) \in L^p_{t,x}, \qquad p > 5/2,$$

and the drift $r^{-1}P_\xi(\nabla r \cdot \nabla\xi)$ and the quadratic $|\nabla\xi|^2\xi$ belong to a Kato class (or are perturbatively small in Campanato). A scalar Hölder theory still does not apply to the system: one would still need ε-regularity for the S^2-flow (Chen–Struwe type) with small $\int|\nabla\xi|^2$, which again uses enstrophy unless a new smallness is found.

(B) Direct Campanato on the angle. For $x \in E(t)$ and ρ small,

$$\rho^{-4}\int_{B_\rho(x)\cap E(t)} 4\sin^2\bigl(\phi(x,y)/2\bigr)\,\mathrm{d}y \leq C(u_0,\nu). \tag{43}$$

The integrand is the Constantin–Fefferman kernel factor. This is Definition 7.1 with no detour through De Giorgi–Nash–Moser.

Energy does not prove (A) or (B). Lipschitz control of ξ is Beale–Kato–Majda. The integral $\int|\nabla\xi|^2$ from Pythagoras needs enstrophy. Lemma 1(B) is the same as (40) via (38).

What remains true. On $\{r > 0\}$ the equation is (28), with the projection and the quadratic term. The stretching force $P_\xi(S\xi)$ lies in $L^2_tL^2_x$ globally by Calderón–Zygmund $2 \to 2$, without $\|\omega\|_{L^\infty}$. That is below the Hölder threshold $p > 5/2$. Definition 7.1 is the only missing analytic ingredient; once (43) holds, Section 11 is unconditional and $T^* = \infty$.

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